Relationships

The following tables give results and formulas useful for understanding and solving the exercises.

Table 1 Definitions of useful functions

rect $\left(\frac{t}{W}\right)$	$\begin{cases} 1 & t \leq \frac{W}{2} \\ 0 & \text{elsewhere} \end{cases}$
triang $\left(\frac{t}{W}\right)$	$\begin{cases} 1 - \frac{\lvert t\rvert}{W/2} & \lvert t\rvert \leq \frac{W}{2} \\ 0 & \text{elsewhere} \end{cases}$
sinc tW	$\frac{\sin \pi t W}{\pi t W}$
sign t	$\begin{cases} -1 & t < 0 \\ 0 & t = 0 \\ 1 & t > 0 \end{cases}$

Table 2 Properties of Fourier transforms

Function	Transform
$ah_1(t) + bh_2(t)$	$aH_1(f) + bH_2(f)$
$h(at)$	$\frac{1}{\lvert a \rvert} H\left(\frac{f}{a}\right)$
$H(t)$	$h(-f)$
$h(t - t_0)$	$H(f)e^{-j2\pi f t_0}$
$h(t)e^{j2\pi f_0 t}$	$H(f - f_0)$
$\frac{dh(t)}{dt}$	$j2\pi f H(f)$
$\int_{-\infty}^{t} h(\tau)\, d\tau$	$\frac{1}{\mathrm{j}2\pi f} H(f) + \frac{H(0)}{2}\delta(f)$
$h^*(t)$	$H^*(-f)$
$h_1(t)h_2(t)$	$\int_{-\infty}^{\infty} H_1(\lambda)H_2(f - \lambda)\, d\lambda$
$\int_{-\infty}^{\infty} h_1(\tau)h_2(t - \tau)\, d\tau$	$H_1(f)H_2(f)$

Table 3 Fourier transform formulas

Function	Transform
$A\ \text{rect}\ \left(\frac{t}{W}\right)$	$AW\ \text{sinc}\ fW$
$A\ \text{sinc}\ Wt$	$\frac{A}{W}\ \text{rect}\ \left(\frac{f}{W}\right)$
1	$\delta(f)$
$\delta(t)$	1
$\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{t^2}{2\sigma^2}}$	$e^{-2\pi^2 f^2 \sigma^2}$
$\sum_{i=-\infty}^{+\infty}\ \delta(t-iT)$	$\frac{1}{T}\ \sum_{i=-\infty}^{+\infty}\ \delta\left(f-\frac{n}{T}\right)$
$\frac{1}{at}$	$-j\ \text{sign}\ f$
$e^{j2\pi f_0 t}$	$\delta(f-f_0)$
$\sin(2\pi f_0 t)$	$\frac{1}{2j}\left[\delta(f-f_0)-\delta(f+f_0)\right]$
$\cos(2\pi f_0 t)$	$\frac{1}{2}\left[\delta(f-f_0)+\delta(f+f_0)\right]$
$\text{triang}\ \left(\frac{t}{W}\right)$	$\frac{W}{2}\ \text{sinc}^2\ \left(\frac{fW}{2}\right)$

Table 4 Results on periodic signals (period =T)

Fourier series expansion	$h(t)=\sum_{-\infty}^{+\infty}\ c_n\ e^{\frac{2\pi n t}{T}}$
Power spectral density	$S_h(f)=\sum_{-\infty}^{+\infty}\ \lvert c_n\rvert^2\ \delta\left(f-\frac{n}{T}\right)$
Power	$P=\frac{1}{T}\int_{-T/2}^{T/2}\ h^2(t)\ dt=\sum_{-\infty}^{+\infty}\ \lvert c_n\rvert^2$
Coefficients c_n	$c_n=\frac{1}{T}\int_{-T/2}^{T/2}\ h(t)\ e^{-j2\pi\frac{nt}{T}}\ dt$
Autocorrelation	$R(\tau)=\frac{1}{T}\int_{-T/2}^{T/2}\ h(t)h(t-\tau)\ dt$ $=\sum_{-\infty}^{+\infty}\ \lvert c_n\rvert^2 e^{\frac{2\pi n\tau}{T}}$

Table 5 Results on aperiodic signals

Fourier Transform	$H(f) = \int_{-\infty}^{\infty} h(t)\, e^{-2\pi f t}\, dt$
Parseval Theorem	$E = \int_{-\infty}^{\infty} h^2(t)\, dt = \int_{-\infty}^{\infty} \|H(f)\|^2\, df$
Autocorrelation	$R(\tau) = \int_{-\infty}^{\infty} h(t)h(t-\tau)\, dt = \mathcal{F}^{-1}(\|H(f)\|^2)$

Table 6 Trigonometric formulas

$\cos\theta$	$\frac{1}{2}\left[e^{j\theta} + e^{-j\theta}\right]$
$\sin\theta$	$\frac{1}{2j}\left[e^{j\theta} - e^{-j\theta}\right]$
$\cos 2\theta$	$\cos^2\theta - \sin^2\theta$
$\sin 2\theta$	$2\sin\theta\cos\theta$
$\cos^2\theta$	$\frac{1}{2}[1 + \cos 2\theta]$
$\sin^2\theta$	$\frac{1}{2}[1 - \cos 2\theta]$
$\sin(\alpha \pm \beta)$	$\sin\alpha\cos\beta \pm \cos\alpha\sin\beta$
$\cos(\alpha \pm \beta)$	$\cos\alpha\cos\beta \mp \sin\alpha\sin\beta$
$\tan(\alpha \pm \beta)$	$\frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$
$\sin\alpha\sin\beta$	$\frac{1}{2}[\cos(\alpha-\beta) - \cos(\alpha+\beta)]$
$\cos\alpha\cos\beta$	$\frac{1}{2}[\cos(\alpha-\beta) + \cos(\alpha+\beta)]$
$\sin\alpha\cos\beta$	$\frac{1}{2}[\sin(\alpha-\beta) + \sin(\alpha+\beta)]$